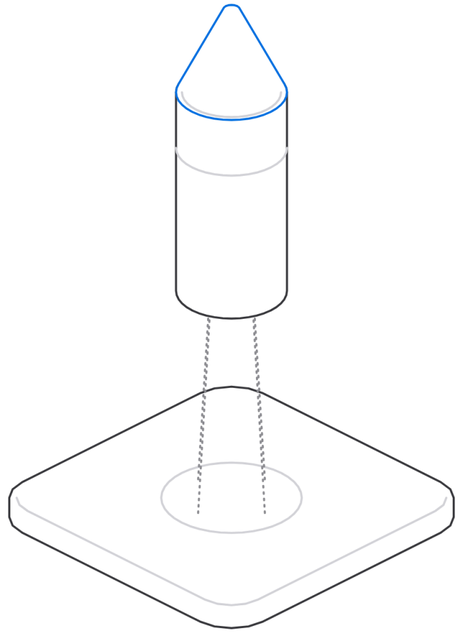
CBSE CLASS 9 · PHYSICS · 60-MINUTE LESSON

Motion and force.

Physics from the very first step: how things move, how we measure it, and why it happens.

UNITS · VECTORS · SPEED AND VELOCITY · ACCELERATION · NEWTON'S LAWS



THE PLAN

Sixty minutes, ten small steps.

01	What is physics?	3–7 min
02	Physical quantities and SI units	7–13 min
03	Scalars and vectors	13–17 min
04	Distance and displacement	17–23 min
05	Speed and velocity	23–30 min
06	Uniform and non-uniform motion	30–35 min
07	Acceleration	35–40 min
08	Equations of motion	40–48 min
09	Force and Newton's laws	48–57 min
10	Practice and recap	57–60 min

Aligned to CBSE Class 9 Science 2026–27 (NCERT *Exploration*: “Describing Motion Around Us” and “How Forces Affect Motion”). Units and vectors are taught as foundations.

HOW WE WILL LEARN

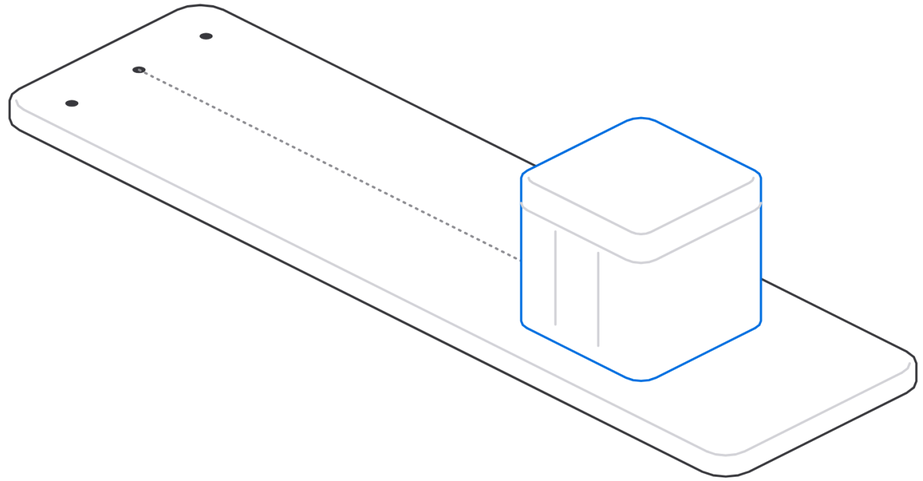
Every idea gets the same four steps.

01 **A simple definition**
In words you could explain to a friend.

02 **An everyday example**
Cars, bicycles, balls and you.

03 **A picture and a formula**
With every symbol and unit explained.

04 **A worked problem**
Step by step, then you try one.



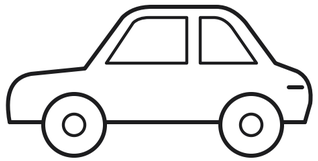
01 · WHAT IS PHYSICS?

Physics is the study of matter, energy, and how they interact.

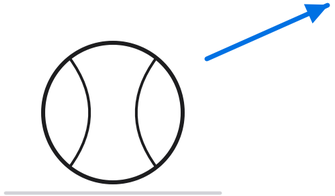
It explains how things move and why. The same few rules work for a bicycle, a cricket ball and a rocket.



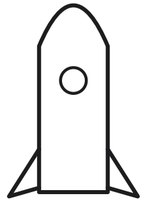
Cycling to school



Braking a car



Bowling a ball



Launching a rocket

MOTION IS EVERYWHERE

Describing motion needs three things.

A reference point

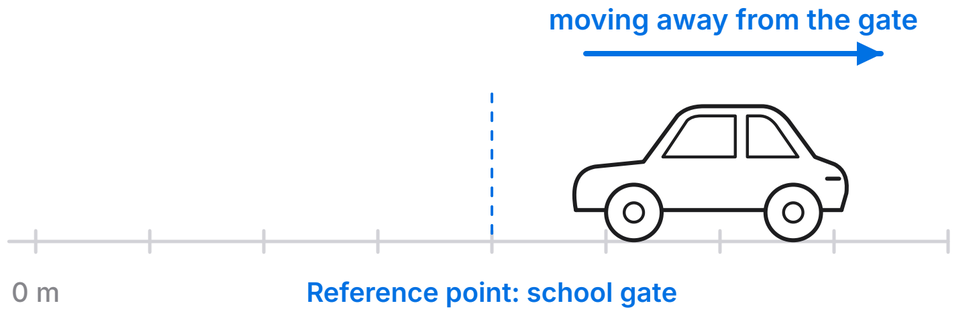
Where are we measuring from? A school gate, a lamp post, the start line.

A measurement

How far, how fast, how long, with a unit.

A direction

Which way? East, up, towards the wall.



ASK THE CLASS

Is a student sitting at a desk moving? Think about the Earth.

02 · PHYSICAL QUANTITIES

A measurement is a number and a unit.

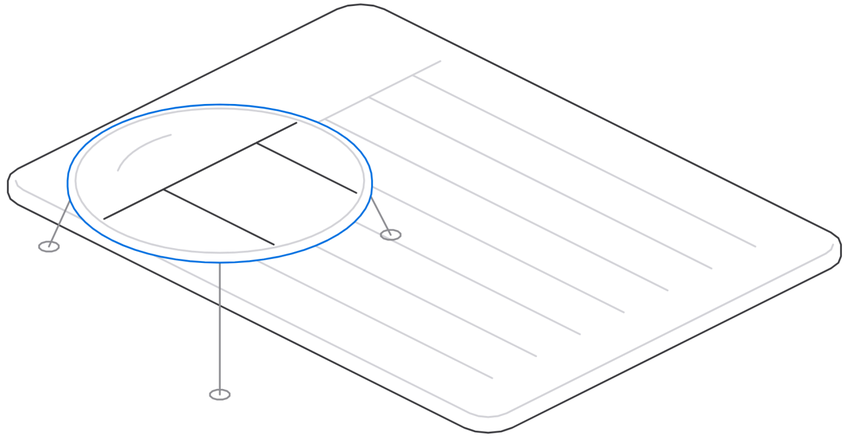
DEFINITION

A **physical quantity** is anything we can measure, such as length, mass or time.

5
means nothing

→

5 m
five metres



THE SI SYSTEM

Seven base units. Everything else is built from them.

SI stands for the International System of Units, used by scientists in every country.

QUANTITY	UNIT	SYMBOL
Length	metre	**m**
Mass	kilogram	**kg**
Time	second	**s**
Electric current	ampere	**A**
Temperature	kelvin	**K**
Amount of substance	mole	**mol**
Luminous intensity	candela	**cd**

TODAY WE NEED THREE

Length in **metres**, mass in **kilograms** and time in **seconds**. Speed, acceleration and force are all built from these.

Watch out: the symbol for second is **s**, not "sec". Units named after people, like ampere, still use a small letter when written in full.

DERIVED UNITS AND PREFIXES

Bigger and smaller units.

PREFIX	SYMBOL	MEANS
kilo	k	× 1000
centi	c	÷ 100
milli	m	÷ 1000

Speed is a **derived** quantity: distance ÷ time. Its SI unit is the **metre per second**, $\mathrm{m/s}$ or $\mathrm{m\ s^{-1}}$.

THE CONVERSION YOU WILL USE MOST

$$1\ \mathrm{km/h} = \frac{1000\ \mathrm{m}}{3600\ \mathrm{s}} = \frac{5}{18}\ \mathrm{m/s}$$

km/h to m/s: multiply by $\frac{5}{18}$. m/s to km/h: multiply by $\frac{18}{5}$.

WORKED EXAMPLE

QUESTION A car moves at 72 km/h. What is this in m/s?

FORMULA $v\ (\mathrm{m/s}) = v\ (\mathrm{km/h}) \times \frac{5}{18}$

SOLVE $72 \times \frac{5}{18} = 4 \times 5$

ANSWER $20\ \mathrm{m/s}$

YOUR TURN

Two quick conversions.

QUESTION 1

A cyclist rides at $18\ \mathrm{km/h}$. What is that in m/s?

QUESTION 2

A cheetah can run at about $30\ \mathrm{m/s}$. What is that in km/h?

03 · SCALARS AND VECTORS

Some quantities need a direction.

SCALAR

Has **size only**. Mass, time, distance, speed, temperature.

VECTOR

Has **size and direction**. Displacement, velocity, acceleration, force.

We draw a vector as an arrow: its length shows the size, its head shows the direction.



"5 km" is a scalar. "5 km north" is a vector.

SORT THEM

Scalar or vector?

Mass
of a school bag

Velocity
of a train going west

Time
taken for a race

Force
pushing a door open

ASK THE CLASS

Hold up one finger for scalar, two for vector, as I point to each card.

04 · DISTANCE AND DISPLACEMENT

How far you went, and how far you got.

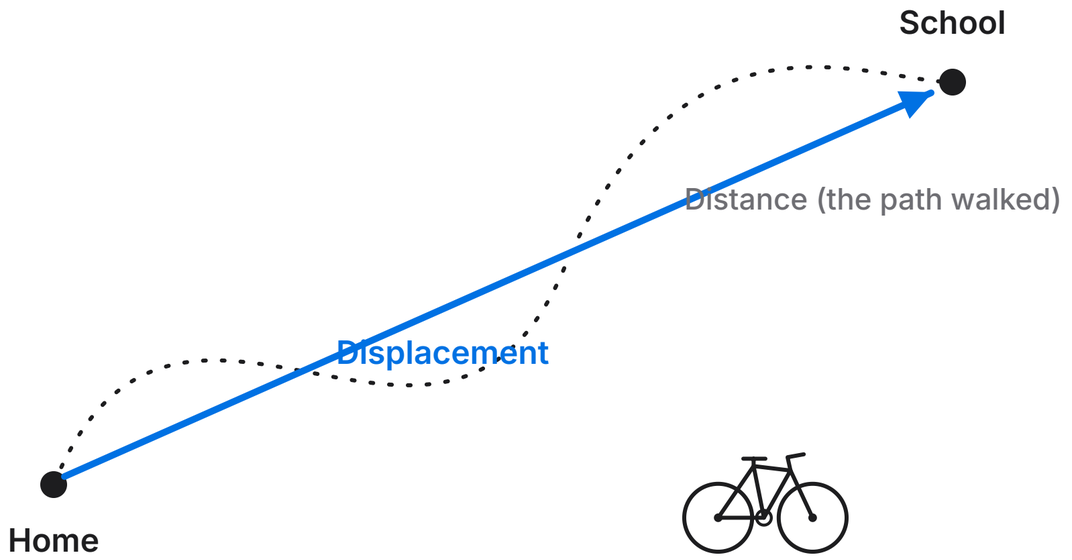
DISTANCE

The **total length of the path** travelled. A scalar.

DISPLACEMENT

The **shortest straight line** from start to finish, with direction. A vector.

Both are measured in metres (m).



WORKED EXAMPLE

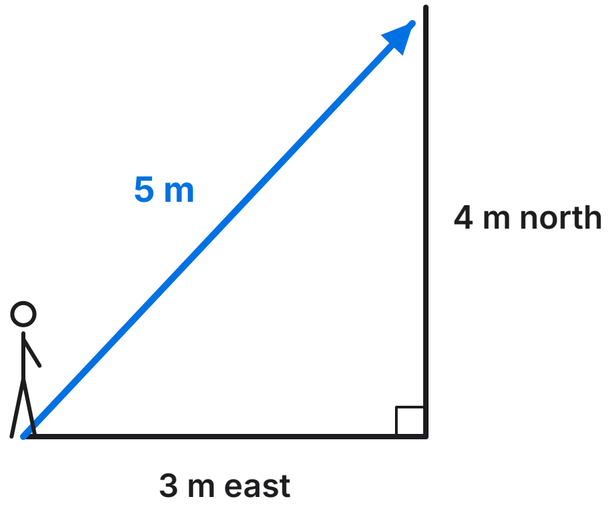
A walk around the playground.

QUESTION — Riya walks 3 m east, then 4 m north. Find the distance and the displacement.

DISTANCE — Add the path lengths: $3 + 4 = 7\ \text{m}$

DISPLACEMENT — The straight line is the hypotenuse: $\sqrt{3^2 + 4^2} = \sqrt{25}$

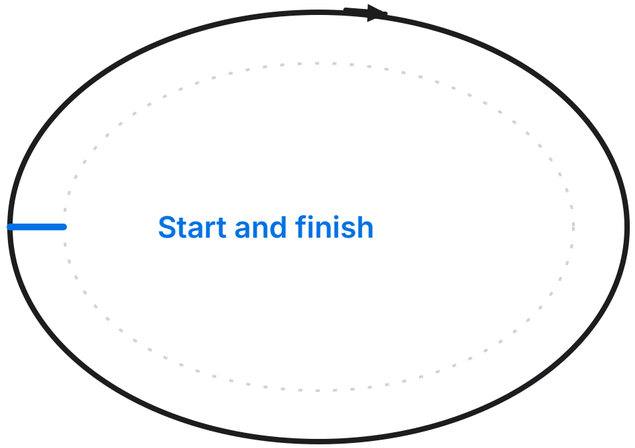
ANSWER — **Distance $= 7\ \text{m}$. Displacement $= 5\ \text{m}$ towards the north-east.**



WORKED EXAMPLE

One lap of the track.

QUESTION	An athlete runs exactly one lap of a 400 m track and stops at the start line.
DISTANCE	The whole lap was run: 400 m.
DISPLACEMENT	Start and finish are the same point, so the straight line between them has zero length.
ANSWER	**Distance** $= 400\ \mathrm{m}$. **Displacement** $= 0\ \mathrm{m}$.



ASK THE CLASS

Can displacement ever be greater than distance?

05 · SPEED

How fast something covers distance.

DEFINITION

Speed is the distance travelled per unit time. A scalar.

$$\text{speed} = \frac{\text{distance}}{\text{time}} \qquad v = \frac{s}{t}$$

SYMBOL	MEANS	SI UNIT
v	speed	m/s
s	distance travelled	m
t	time taken	s



A speedometer shows the speed at each instant.

VELOCITY

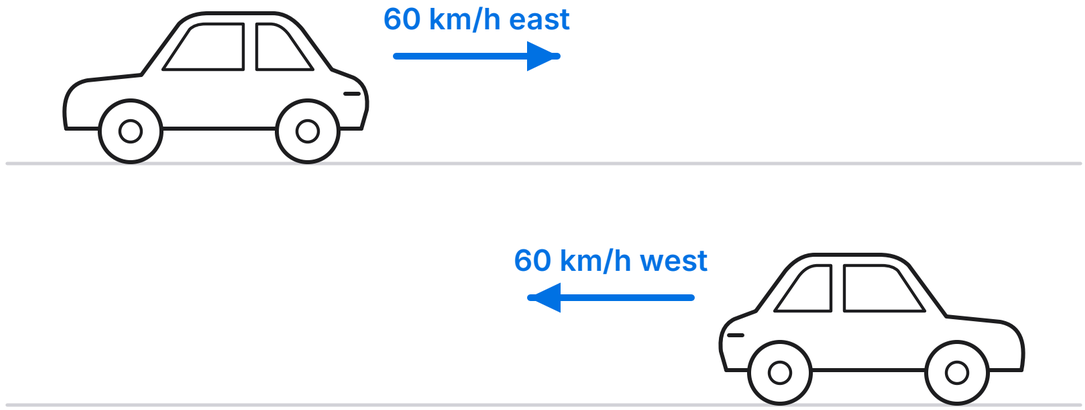
Speed with a direction.

DEFINITION

Velocity is the displacement per unit time. A vector.

$$\text{velocity} = \frac{\text{displacement}}{\text{time}}$$

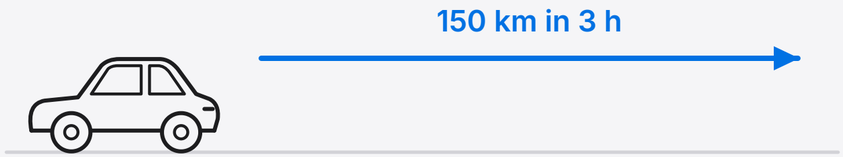
Same SI unit as speed: **m/s**. But the answer must say which way.



Same speed. Different velocities, because the directions differ.

WORKED EXAMPLES

Speed and velocity in numbers.

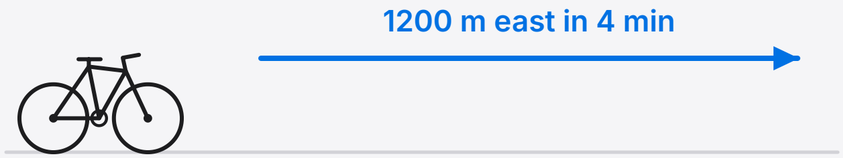


QUESTION — A car covers 150 km in 3 h. Find its average speed.

FORMULA — $v = \frac{s}{t}$

SOLVE — $\frac{150\ \text{km}}{3\ \text{h}} = 50\ \text{km/h}$

ANSWER — $50\ \text{km/h}$ **or about** $13.9\ \text{m/s}$



QUESTION — A cyclist rides 1200 m due east in 4 minutes. Find her velocity.

CONVERT — $4\ \text{min} = 4 \times 60 = 240\ \text{s}$

SOLVE — $\frac{1200\ \text{m}}{240\ \text{s}} = 5\ \text{m/s}$

ANSWER — $5\ \text{m/s}$ **due east**

AVERAGE SPEED

The trap almost everyone falls into.

$$\text{average speed} = \frac{\text{total distance}}{\text{total time}}$$

QUESTION	Go 60 km at 60 km/h. Return the same 60 km at 30 km/h. Average speed?
TIMES	Going: 1 h. Returning: 2 h. Total: 3 h.
SOLVE	$\frac{120\ \text{km}}{3\ \text{h}}$
ANSWER	$40\ \text{km/h}$ **(not 45 km/h)**

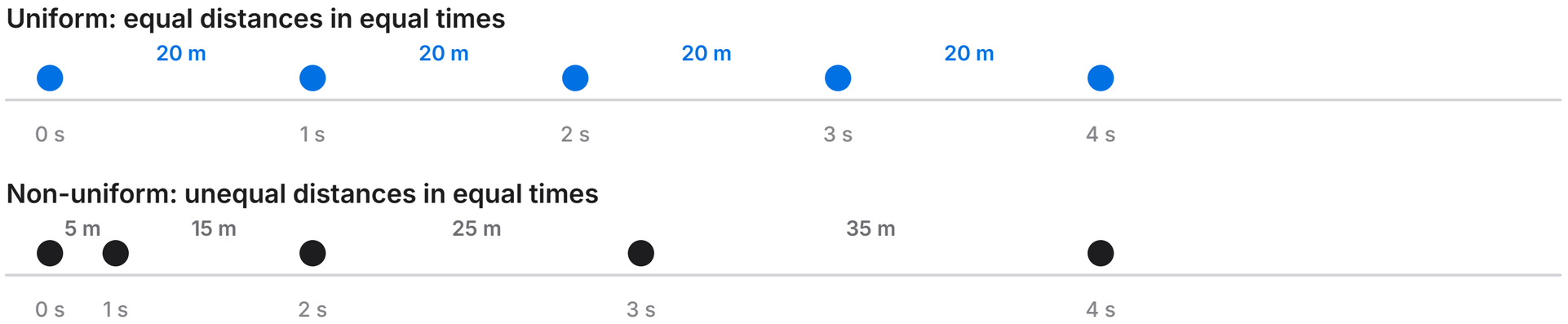
ASK THE CLASS

Why is the answer not simply the average of 60 and 30?

Remember: never average the speeds. Always divide total distance by total time. The slow part takes longer, so it counts for more.

06 · UNIFORM AND NON-UNIFORM MOTION

Is the speed steady, or changing?



UNIFORM MOTION

Equal distances in equal intervals of time. A car on cruise control.

NON-UNIFORM MOTION

Unequal distances in equal intervals. A bus leaving a stop.

DISTANCE–TIME GRAPHS

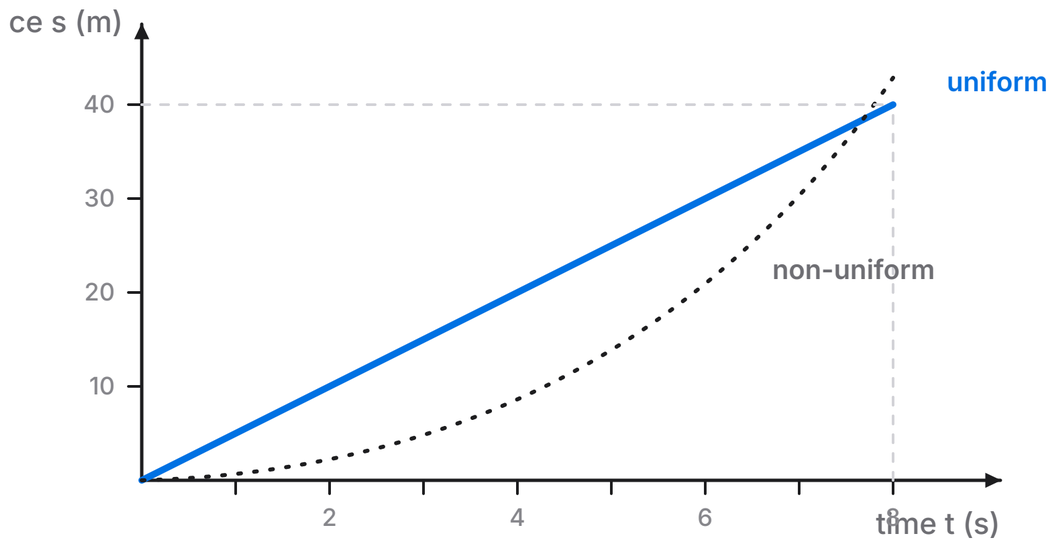
The slope tells you the speed.

A straight line means **uniform** speed. A curve means the speed is **changing**. The steeper the line, the faster the object.

READ — The blue line reaches 40 m at 8 s.

SOLVE — $v = \frac{40\ \text{m}}{8\ \text{s}}$

ANSWER — $5\ \text{m/s}$



07 · ACCELERATION

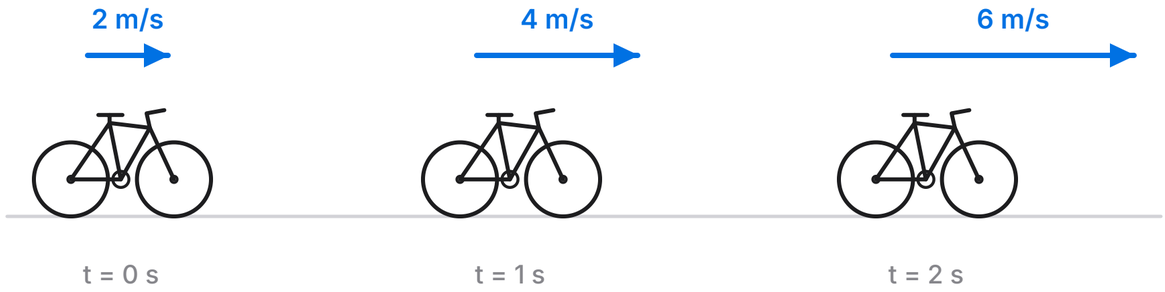
How quickly velocity changes.

DEFINITION

Acceleration is the change in velocity per unit time. A vector.

$$a = \frac{v - u}{t}$$

SYMBOL	MEANS	SI UNIT
a	acceleration	m/s²
u	initial velocity	m/s
v	final velocity	m/s
t	time taken	s



Velocity rises by 2 m/s every second: $a = 2\ \mathrm{m/s^2}$

WORKED EXAMPLES

Speeding up and slowing down.

QUESTION	A car starts from rest and reaches 20 m/s in 5 s. Find its acceleration.
GIVEN	$u = 0,\ v = 20\ \text{m/s},\ t = 5\ \text{s}$
FORMULA	$a = \frac{v-u}{t}$
SOLVE	$\frac{20-0}{5}$
ANSWER	$a = 4\ \text{m/s}^2$

QUESTION	A cyclist brakes from 6 m/s to rest in 3 s. Find the acceleration.
GIVEN	$u = 6\ \text{m/s},\ v = 0,\ t = 3\ \text{s}$
FORMULA	$a = \frac{v-u}{t}$
SOLVE	$\frac{0-6}{3}$
ANSWER	$a = -2\ \text{m/s}^2$, **slowing down**

ASK THE CLASS

What does the minus sign in the second answer tell us?

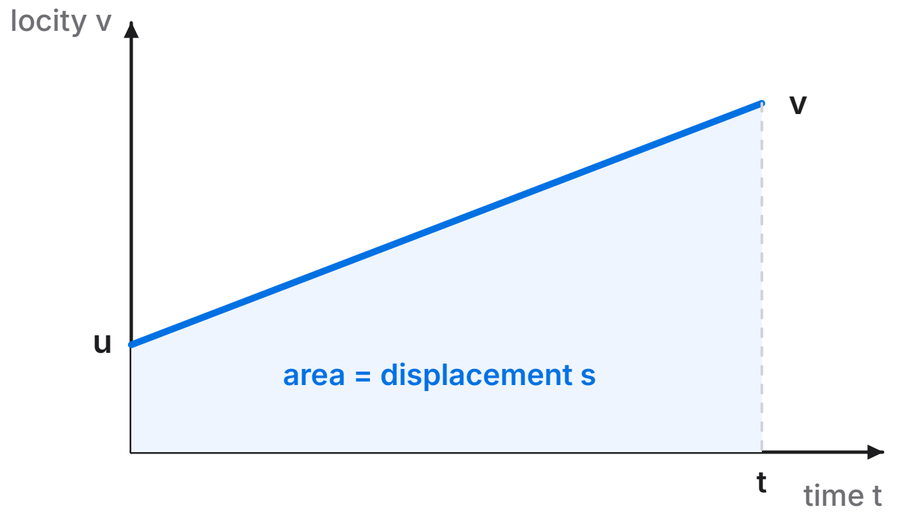
08 · EQUATIONS OF MOTION

Three equations for uniform acceleration.

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$



On a velocity–time graph, the slope is a and the area under the line is s.

They work only when the acceleration is **uniform** (constant) and the motion is in a straight line.

KNOW YOUR LETTERS

Five quantities. Each equation leaves one out.

SYMBOL	MEANS	SI UNIT
u	initial velocity	m/s
v	final velocity	m/s
a	acceleration	m/s²
t	time	s
s	displacement	m

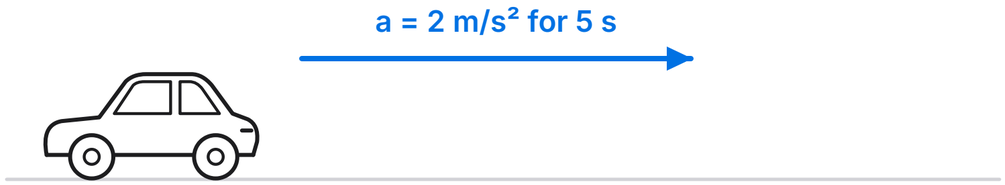
EQUATION	LEAVES OUT
$v = u + at$	s
$s = ut + \frac{1}{2}at^2$	v
$v^2 = u^2 + 2as$	t

THE METHOD

1. Write what you know. 2. Write what you need. 3. Pick the equation that does not use the one quantity you neither know nor need.

WORKED EXAMPLE 1

A car pulls away.



A car starts from rest and accelerates uniformly at $2\ \mathrm{m/s^2}$ for $5\ \mathrm{s}$. Find its final velocity and the distance covered.

GIVEN	$u = 0,\ a = 2\ \mathrm{m/s^2},\ t = 5\ \mathrm{s}$
VELOCITY	$v = u + at = 0 + 2 \times 5 = 10\ \mathrm{m/s}$
DISTANCE	$s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 2 \times 5^2$
	$= 25\ \mathrm{m}$
CHECK	$v^2 = 2as \Rightarrow 100 = 2 \times 2 \times 25$ ✓
ANSWER	$v = 10\ \mathrm{m/s},\ s = 25\ \mathrm{m}$

WORKED EXAMPLE 2

A train brakes.

A train moving at $20\ \mathrm{m/s}$ brakes with a uniform acceleration of $-0.5\ \mathrm{m/s^2}$. How far does it travel before stopping, and how long does it take?

ASK THE CLASS

Which equation finds the distance without knowing the time?

GIVEN	$u = 20\ \mathrm{m/s},\ v = 0,\ a = -0.5\ \mathrm{m/s^2}$
DISTANCE	$v^2 = u^2 + 2as \Rightarrow 0 = 400 + 2(-0.5)s$
	$s = \frac{400}{1} = 400\ \mathrm{m}$
TIME	$v = u + at \Rightarrow 0 = 20 - 0.5t$
	$t = \frac{20}{0.5} = 40\ \mathrm{s}$
ANSWER	$s = 400\ \mathrm{m},\ t = 40\ \mathrm{s}$

09 · FORCE

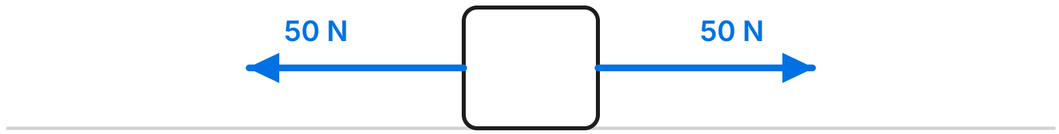
A push or a pull.

DEFINITION

A **force** is a push or pull that can change an object's **speed**, **direction** or **shape**. A vector, measured in **newtons (N)**.

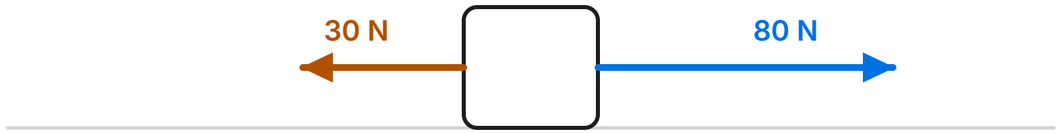
Only an **unbalanced** (net) force changes motion.

Balanced: equal and opposite



no change in motion

Unbalanced: one side wins

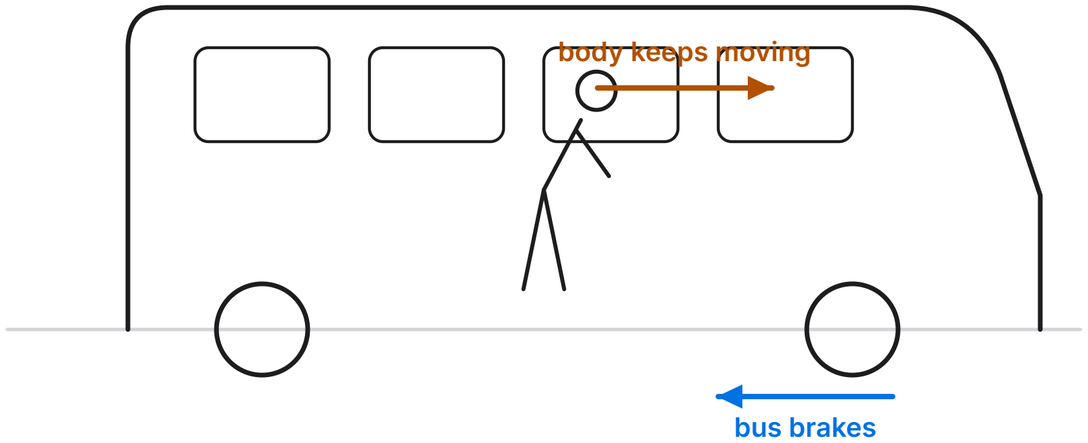


NEWTON'S FIRST LAW

Objects keep doing what they are doing.

> FIRST LAW (LAW OF INERTIA)
>
> An object stays at rest, or keeps moving in a straight line at a steady speed, unless an unbalanced force acts on it.

Inertia is this tendency to resist a change in motion. More mass means more inertia.



When the bus brakes, your body tends to keep moving forwards.

NEWTON'S SECOND LAW

Force, mass and acceleration.

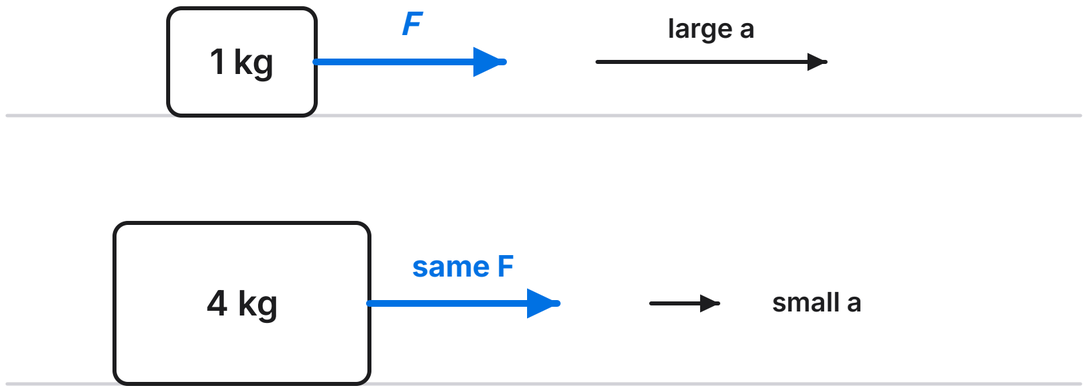
SECOND LAW

The acceleration of an object is directly proportional to the net force and inversely proportional to its mass.

$$F = ma$$

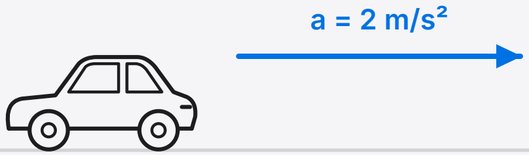
SYMBOL	MEANS	SI UNIT
F	net force	newton (N)
m	mass	kg
a	acceleration	m/s²

$1\ \mathrm{N} = 1\ \mathrm{kg\ m/s^2}$



WORKED EXAMPLES

Using F = ma.

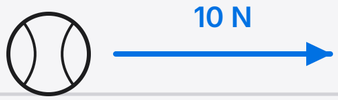


QUESTION — What force gives a 1000 kg car an acceleration of 2 m/s²?

FORMULA — $F = ma$

SOLVE — 1000×2

ANSWER — $F = 2000\ \text{N}$



QUESTION — A net force of 10 N acts on a 0.5 kg ball. Find its acceleration.

FORMULA — $a = \frac{F}{m}$

SOLVE — $\frac{10}{0.5}$

ANSWER — $a = 20\ \text{m/s}^2$

NEWTON'S THIRD LAW

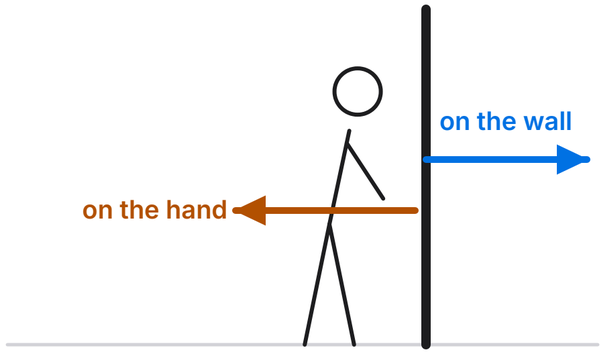
Forces come in pairs.

THIRD LAW

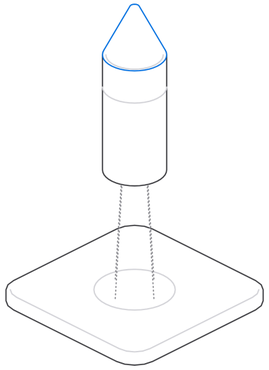
For every action there is an equal and opposite reaction.

The two forces act on **different objects**, so they never cancel each other out.

Walking: your foot pushes the ground back, the ground pushes you forward. Swimming, a gun's recoil and a rocket all work the same way.



Two objects, two forces



Gases pushed down, rocket pushed up

YOUR TURN

Which law is it?

SITUATION A

A tablecloth is pulled quickly and the dishes stay on the table.

SITUATION B

Kicking a football harder makes it speed up more.

SITUATION C

A swimmer pushes water backwards and moves forwards.

PRACTICE

CBSE-style questions.

Try Q1 and Q2 now. The rest are for homework; full solutions follow.

Q1 CONCEPTUAL

Can an object have zero displacement but a non-zero distance? Give an example.

Q2 NUMERICAL

Convert 54 km/h into m/s, and 25 m/s into km/h.

Q3 NUMERICAL

A bus starts from rest and accelerates uniformly at 0.1 m/s^2 for 2 minutes. Find its final speed and the distance it travels.

Q4 NUMERICAL

A train moving at 90 km/h brakes with a uniform acceleration of -0.5 m/s^2. How far does it travel before it stops?

PRACTICE

CBSE-style questions, continued.

Q5 NUMERICAL

An athlete runs round a circular track of diameter 200 m and completes one round in 40 s. Find the distance and the displacement after 2 min 20 s. (Take π = 22/7.)

Q6 NUMERICAL

What net force is needed to give a 1500 kg car an acceleration of 2 m/s²?

Q7 CONCEPTUAL

Passengers in a bus fall forward when it stops suddenly. Explain why, using Newton's first law.

Q8 CONCEPTUAL

Action and reaction forces are equal and opposite. Explain why they do not cancel each other.

ANSWER KEY

Detailed solutions, Q1 to Q4.

Q1 Yes. Distance is the length of the path, displacement is the straight line from start to finish. An athlete who runs one full lap of a track covers a distance of 400 m, but ends at the start, so the displacement is 0.

Q2 $54 \times \frac{5}{18} = 15 \text{ m/s}$. $25 \times \frac{18}{5} = 90 \text{ km/h}$.

Q3 $u = 0,\ a = 0.1 \text{ m/s}^2,\ t = 2 \text{ min} = 120 \text{ s}$. $v = u + at = 0.1 \times 120 = 12 \text{ m/s}$. $s = \frac{1}{2}at^2 = \frac{1}{2} \times 0.1 \times 120^2 = 720 \text{ m}$.

Q4 $u = 90 \times \frac{5}{18} = 25 \text{ m/s}, v = 0, a = -0.5 \text{ m/s}^2$. $v^2 = u^2 + 2as \Rightarrow 0 = 625 - s$, so $s = 625 \text{ m}$.

ANSWER KEY

Detailed solutions, Q5 to Q8.

Q5 $2\ \text{min}\ 20\ \text{s} = 140\ \text{s}$, so rounds $= \frac{140}{40} = 3.5$. Circumference $= \pi d = \frac{22}{7} \times 200 = \frac{4400}{7}\ \text{m}$. Distance $= 3.5 \times \frac{4400}{7} = 2200\ \text{m}$. After 3.5 rounds the athlete is on the opposite side, so displacement $= d = 200\ \text{m}$.

Q6 $F = ma = 1500 \times 2 = 3000\ \text{N}$.

Q7 The bus stops because the brakes act on it, but the passengers' bodies tend to stay in motion because of inertia (Newton's first law), so they move forward.

Q8 Action and reaction act on two different objects. Forces can cancel only when they act on the same object, so the pair does not cancel.

RECAP

Nine ideas to take home.

01
Physical quantity
a number and a unit

02
Scalar / vector
size only / size and direction

03
Distance / displacement
path length / straight line, with direction

04
Speed / velocity
distance per time / displacement per time

05
Uniform motion
equal distances in equal times

06
Acceleration
change in velocity per unit time

07
First law
inertia: no change without a net force

08
Second law
F = ma

09
Third law
equal and opposite pairs, on different objects

FORMULA SHEET

Everything you need for today's questions.

$$v = \frac{s}{t}$$

speed

$$\text{avg speed} = \frac{\text{total distance}}{\text{total time}}$$

average speed

$$a = \frac{v - u}{t}$$

acceleration

$$v = u + at$$

first equation

$$s = ut + \frac{1}{2}at^2$$

second equation

$$v^2 = u^2 + 2as$$

third equation

$$F = ma$$

Newton's second law

$$1\ \text{km/h} = \frac{5}{18}\ \text{m/s}$$

conversion

Equations of motion apply only to uniform acceleration in a straight line. Always use SI units: m, kg, s.

NEXT LESSON

Motion in graphs, and circular motion.

Homework: practice questions 3 to 8. Bring your answers; we will check them together using the answer key.

